Appendix 2: Debt Sustainability Analysis Update

1. **This debt sustainability analysis (LIC-DSA) updates the assessment included in the HIPC completion point documentation,[1] using the forward-looking debt sustainability framework (DSF) for low-income countries.** The present analysis confirms the conclusion of the previous DSA that, despite significant debt relief, Comoros remains at high risk of debt distress and debt dynamics remain vulnerable to a number of shocks over the projection period.

2. **Comoros reached the decision point of the HIPC Initiative in July 2010 and was granted debt relief by a number of multilateral and bilateral creditors, following the completion point in December 2012.** The authorities have continued making good faith efforts toward completing agreements with Paris Club and non-Paris Club bilateral creditors, as well as other official and commercial creditors on the provision of debt relief on terms compatible with the relief accorded under the November 2009 Paris Club Agreement and in the context of the HIPC Initiative. Understandings have now been reached with all the major non-Paris Club creditors, representing 16 percent of the total nominal stock of debt at end-2011. This includes, the Abu Dhabi Fund for Development, the Arab Bank for Development in Africa, the Islamic Development Bank, the Kuwait Fund, the Saudi Fund for Development, and Mauritius. One new loan ($41.6 million) was contracted in 2013 for project financing, but fell slightly below the concessionality benchmark established under the ECF.

Text Table 1. Nominal of Stock of External Debt
(Millions of U.S. dollars; end-of-period)

	2012 [1]	2013 [2]
Total external debt	**130.9**	**172.3**
Multilateral creditors	**96.7**	**96.6**
IMF	14.7	14.7
IDA	10.7	10.6
AfDB	0.0	0.0
BADEA	28.3	28.3
Other multilateral creditors	43.1	43.0
Official Bilaterals	**32.5**	**74.0**
Paris Club creditors	5.0	5.0
Non-Paris club creditors	27.6	69.1
UAE	0.0	0.0
Mauritius	0.6	0.5
Kuwait	27.0	27.0
India [3]	0.0	41.6
Commercial creditors	**1.6**	**1.6**

[1] After HIPC and MDRI relief. Part of the debt was still under restructuring/negotiation.
[2] Following Paris Club cancellation of all of its HIPC-eligible deb, rescheduling of short-term debt in arrears, and restructuring non-Paris club debt.
[3] Disbursements are expected over a 4-year period, starting in 2014.

[1] *Union of the Comoros—Enhanced Initiative for Heavily Indebted Poor Countries—Completion Point Document and Multilateral Debt Relief Initiative (*EBS/12/155, 12/03/12).

3. **The present value of the stock of debt at 22.4 percent of GDP slightly exceeds that in the last DSA prepared in December 2012, reflecting some depreciation of the exchange rate, and new lending.** The DSA takes into account all debt relief granted to date by multilateral organizations, Paris Club creditors, and the current status of relief provided by non-Paris Club creditors. The DSA also adds arrears that are now falling due but were not included in previous estimates. Also adding to the debt profile are the remaining credits and loans with Paris Club creditors of original maturity of one year or less for which the cancellation rate was only 47 percent. Finally, the new assessment is based on a discount rate of 5 percent, in line with the approved changes to the framework.

4. **The baseline macroeconomic assumptions for the current DSA are consistent with the macroeconomic framework underlying the current ECF arrangement** (see main document tables 1 and 4) and remain unchanged from the previous assessment. Real GDP growth is projected to accelerate over the medium term to 4 percent per year, CPI inflation over the long-term is projected to remain stable at around 3 percent per year, the current account deficit (including official grants) is expected to widen in 2014 on account of lower exports reflecting decline in commodity prices, and higher import volume associated with capital spending and projected increase in import prices for food and energy products, before it gradually declines, converging to about 7 percent of GDP over the projection period, remittances and FDI are projected to remain robust, fiscal consolidation is expected to continue over the medium term, and financing are assumed to be covered initially assumed mostly on grant terms, followed by loan financing for which the concessionality level gradually drops during the projection period. Strong fiscal adjustment will be required for the present DSA projections to materialize.

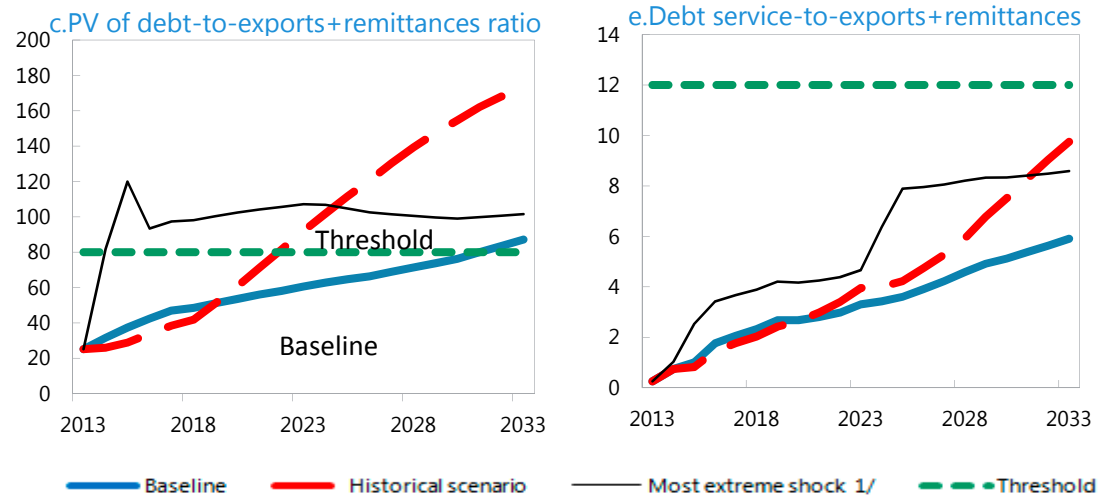
Baseline

5. **Under the baseline scenario, most external debt indicators for Comoros are projected to remain below their relevant indicative thresholds over the medium term** (Table 1a, Figure 1a). The present value of debt to GDP and revenues and debt service indicators are projected to remain below critical benchmarks over the projection span. However, compared to the external debt indicators in the last analysis, the current analysis shows some deterioration. Specifically, the debt to exports ratio is increasing at a faster pace, breaching the critical benchmark by the end of this decade, reflecting more conservative assumptions about export growth. The present value (PV) of public and publicly guaranteed (PPG) external debt to exports ratio is projected to be about 65 percent in 2013, below the critical benchmark, and gradually increases to above the 100 percent threshold for most of the projection period. In addition to the base effect, debt service is higher

compared to the previous analysis, reflecting the service falling due on arrears that had been assumed to be cancelled, but were not part of the final debt relief.

6. **Relevant debt ratios are substantially better when including remittances** (Text Figure 1). Including remittances, which account for some 20 percent of GDP and have been remarkably stable in recent years, the present value of external debt to exports is projected to be substantially lower than in the base case at around 25 percent in 2013, and although gradually increasing, remains below the 80 percent threshold over most of the projection period. The ratio of debt service to exports (plus remittances) similarly shows a substantially better picture.

Text Figure 1. Comoros: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios Including Remittances, 2013-2033 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2023. In figure b. it corresponds to a Combination shock; in c. to a Non-debt flows shock; in d. to a Combination shock; in e. to a Non-debt flows shock and in figure f. to a Combination shock

Sensitivity Analysis

7. **The external debt outlook remains vulnerable to shocks, especially shocks to non-debt creating flows (e.g. FDI) and a combination of several other shocks** (Table 1b, and Figure 1b). The policy thresholds for three key ratios are breached under the most extreme stress tests for a prolonged period of time. Both the PV of external debt-to-GDP and the PV of external debt-to-revenues indicators deteriorate significantly under the scenario assuming a combined shock and a shock to non-debt creating flows.[2] Both indicators are most vulnerable to a combined-shock scenario entailing weaker GDP growth, lower exports, and subdued non-debt creating flows.

[2] In light of the exceptional amount of grants in 2013, the analysis assumes a smaller grant amount, as projected in 2014, which stabilizes the primary deficit at 1.1 percent of GDP over the projection span.

8. **The inclusion of Comoros' domestic public debt in the analysis does not significantly alter the dynamics of debt burden indicators** (Table 2a, 2b). Under the baseline scenario, the PV of total public debt-to-GDP is projected to increase moderately, but remain around 30 percent of GDP, over the medium term, reaching 35 percent by the end of the projection span. However, the present value of total public debt-to-revenue (including grants) ratio is projected to increase steadily over time, reaching rather high levels (82 percent) over the medium term. Further, the evolution of public debt indicators is sensitive to a wide range of shocks (Table 2b). The dynamics of debt appears unsustainable when key variables are kept at historical levels. While no full information on contingent liabilities is available, based on partial data, it seems that their effects are more than adequately covered in the effects of a 10 percent shock to debt creating flows in 2014.

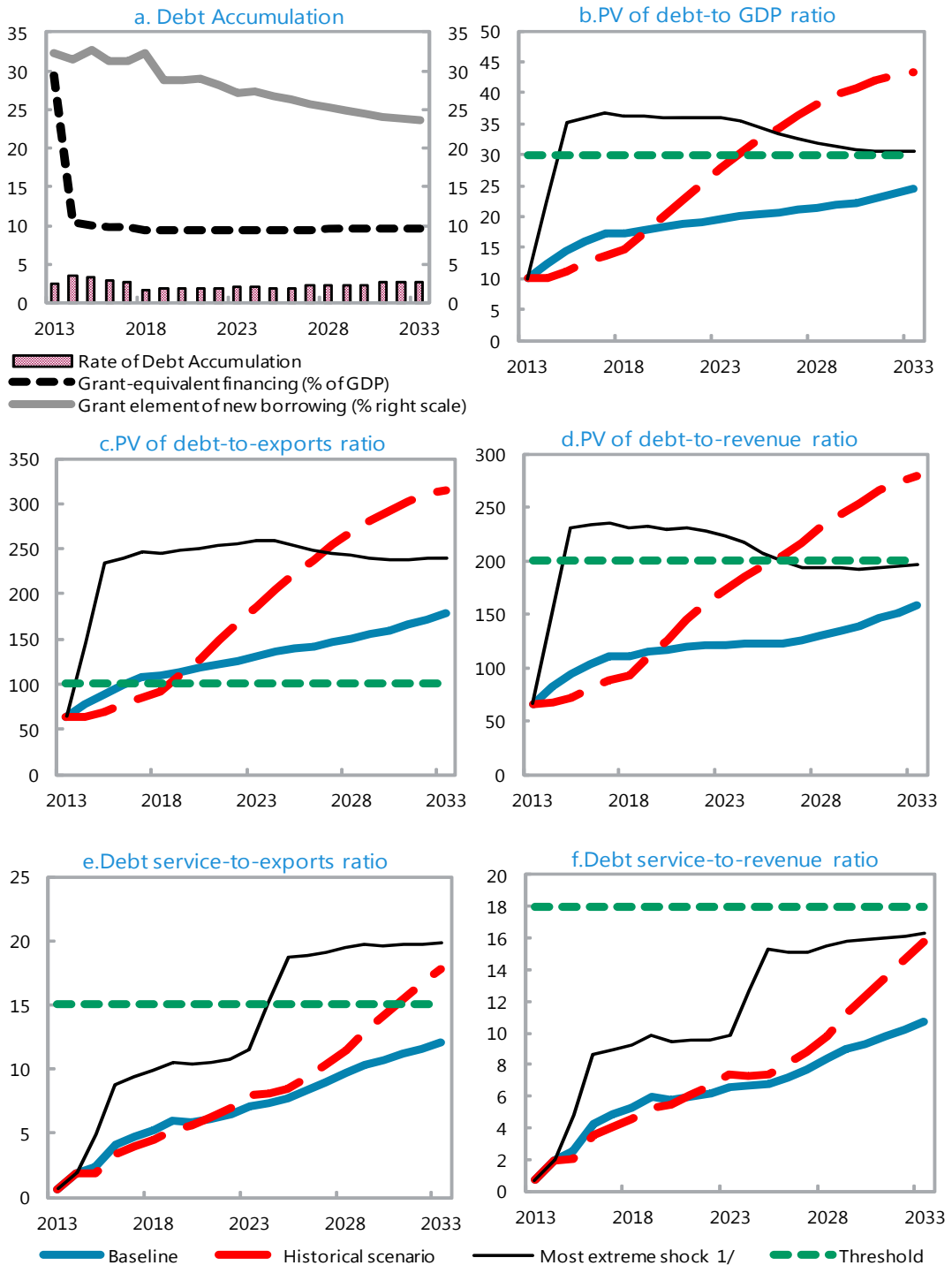
9. **The results of the assessment highlight the paramount importance of continued efforts in fiscal consolidation and structural reforms to achieve higher growth and ensure debt sustainability.** To that end, more efforts to diversify the economy and increase the scope for private sector contribution to key sectors, underpinned by steady progress in developmental and capacity building, will help sustain the growth momentum going forward.

Conclusion

10. **The DSA indicates that Comoros remains at high risk of debt distress, despite debt relief following the HIPC completion point in December 2012.** Under the baseline scenario the debt ratio and debt service indicators are rising gradually over the medium-term, increasing the risk of breaching critical benchmarks. Further, the stress tests indicate serious risks to sustainability.

11. **Ensuring sustainability and improving the external debt outlook will depend on maintaining sound macroeconomic policies.** The inclusion of Comoros' domestic debt in the analysis reinforces the conclusions of the external DSA and stresses the risks to debt prospects absent prudent policies to build the necessary buffers to be able to weather the effects of external shocks. In this context, it is essential that the authorities continue current efforts to strengthen public finance management, reform state-owned enterprises, and improve the investment climate so as to lay the foundation for rapid sustainable economic growth and ensure robust debt sustainability in the face of volatile shocks.

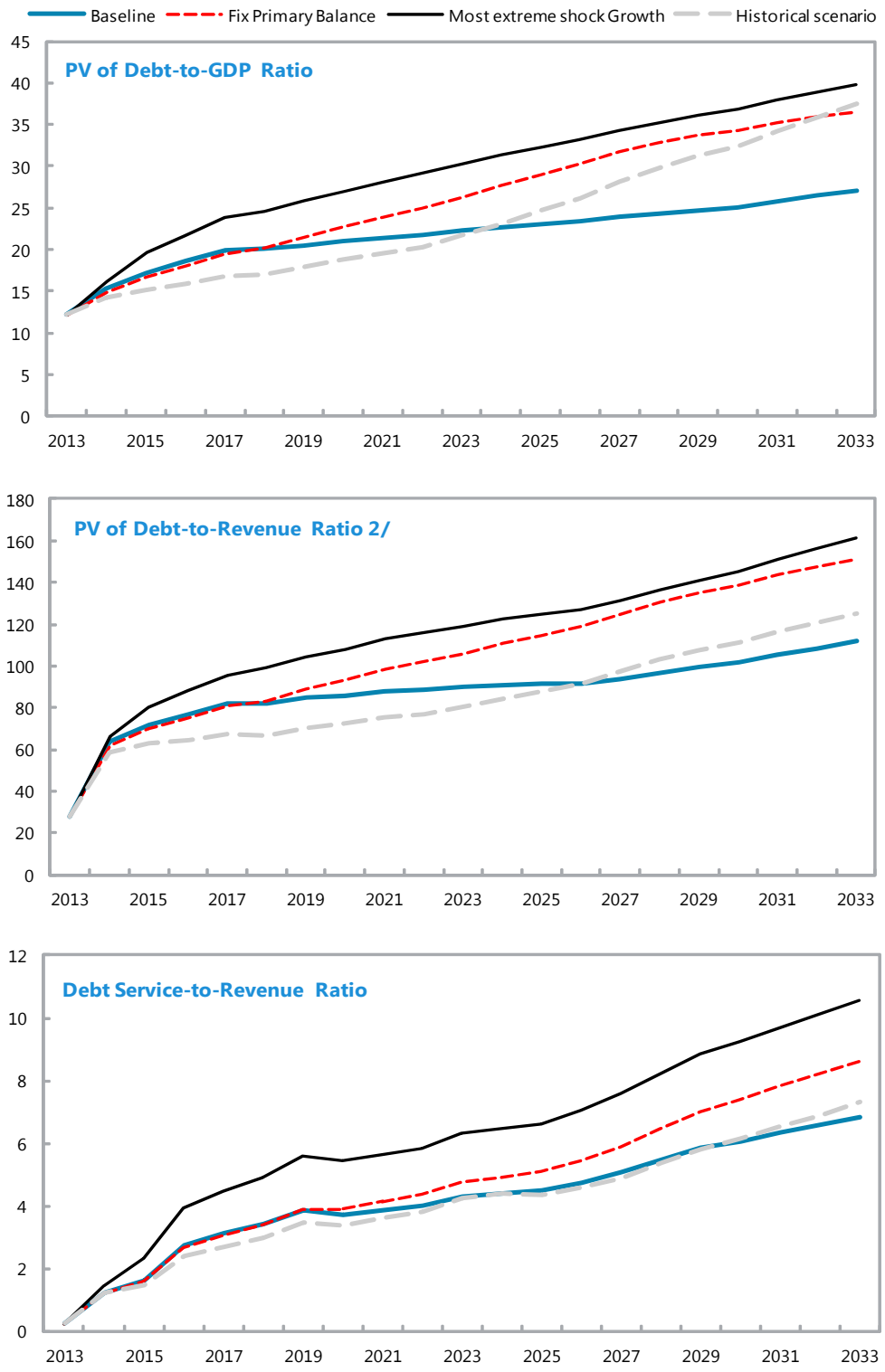
Figure 1a. Comoros: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2013-2033 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2023. In figure b. it corresponds to a Combination shock; in c. to a Combination shock; in d. to a Combination shock; in e. to a Combination shock and in figure f. to a Combination shock

Figure 1b. Comoros: Indicators of Public Debt Under Alternative Scenarios, 2013-2033 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2023.
2/ Revenues are defined inclusive of grants.

Table 1a.: External Debt Sustainability Framework, Baseline Scenario, 2010-2033 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2010	2011	2012			2013	2014	2015	2016	2017	2018	2013-2018 Average	2023	2033	2019-2033 Average
External debt (nominal) 1/	**49.2**	**44.7**	**18.9**			**20.2**	**22.7**	**25.1**	**26.7**	**27.9**	**27.7**		**28.6**	**32.6**	
of which: public and publicly guaranteed (PPG)	49.2	44.7	18.9			20.2	22.7	25.1	26.7	27.9	27.7		28.6	32.6	
Change in external debt	-2.7	-4.5	-25.8			1.3	2.5	2.3	1.7	1.2	-0.2		0.4	0.7	
Identified net debt-creating flows	3.4	0.3	3.2			2.9	7.3	6.0	4.5	4.4	3.6		0.8	2.5	
Non-interest current account deficit	**5.3**	**9.1**	**3.5**	**6.2**	**2.7**	**6.0**	**10.4**	**8.8**	**7.1**	**6.9**	**5.9**		**2.5**	**3.7**	2.9
Deficit in balance of goods and services	34.2	33.9	39.0			37.5	36.5	33.6	31.1	29.2	27.8		22.6	20.4	
Exports	15.7	16.2	14.9			15.6	15.8	16.1	16.2	16.1	15.9		14.9	13.7	
Imports	49.9	50.2	53.9			53.1	52.3	49.7	47.3	45.3	43.7		37.5	34.1	
Net current transfers (negative = inflow)	-28.9	-24.9	-35.4	-21.5	7.5	-31.5	-26.2	-24.9	-24.0	-22.2	-21.6		-20.0	-16.9	-19.1
of which: official	-8.9	0.0	0.0			-2.8	1.0	1.3	1.2	1.2	1.2		0.3	0.3	
Other current account flows (negative = net inflow)	0.0	0.0	-0.2			0.0	0.1	0.1	0.0	-0.1	-0.3		-0.1	0.2	
Net FDI (negative = inflow)	**-1.5**	**-3.8**	**-1.7**	**-1.3**	**1.2**	**-2.6**	**-2.4**	**-2.3**	**-2.1**	**-1.9**	**-1.8**		**-1.1**	**-0.5**	-0.9
Endogenous debt dynamics 2/	**-0.3**	**-5.0**	**1.5**			**-0.5**	**-0.6**	**-0.6**	**-0.6**	**-0.6**	**-0.5**		**-0.5**	**-0.7**	
Contribution from nominal interest rate	0.4	0.3	0.4			0.1	0.2	0.3	0.3	0.4	0.5		0.6	0.8	
Contribution from real GDP growth	-1.0	-1.0	-1.4			-0.6	-0.7	-0.8	-0.9	-1.0	-1.1		-1.2	-1.5	
Contribution from price and exchange rate changes	0.3	-4.4	2.5			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-6.1**	**-4.8**	**-29.0**			**-1.5**	**-4.8**	**-3.7**	**-2.8**	**-3.2**	**-3.8**		**-0.5**	**-1.8**	
of which: exceptional financing	...	0.0	-22.6			...	...	...	...	...	...		...	...	
PV of external debt 4/	...	...	8.4			10.0	12.3	14.4	16.0	17.2	17.3		19.6	24.4	
In percent of exports	...	...	56.5			64.1	78.0	89.2	98.4	107.0	109.2		131.1	177.8	
PV of PPG external debt	**...**	**...**	**8.4**			**10.0**	**12.3**	**14.4**	**16.0**	**17.2**	**17.3**		**19.6**	**24.4**	
In percent of exports	**...**	**...**	**56.5**			**64.1**	**78.0**	**89.2**	**98.4**	**107.0**	**109.2**		**131.1**	**177.8**	
In percent of government revenues	**...**	**...**	**43.6**			**65.9**	**82.1**	**94.4**	**103.5**	**110.5**	**110.6**		**121.3**	**157.5**	
Debt service-to-exports ratio (in percent)	**11.4**	**10.0**	**10.6**			**0.7**	**1.8**	**2.4**	**4.1**	**4.7**	**5.2**		**7.1**	**12.1**	
PPG debt service-to-exports ratio (in percent)	**11.4**	**10.0**	**10.6**			**0.7**	**1.8**	**2.4**	**4.1**	**4.7**	**5.2**		**7.1**	**12.1**	
PPG debt service-to-revenue ratio (in percent)	**12.5**	**10.1**	**8.2**			**0.7**	**1.9**	**2.5**	**4.3**	**4.8**	**5.3**		**6.6**	**10.7**	
Total gross financing need (Billions of U.S. dollars)	0.0	0.0	0.0			0.0	0.1	0.1	0.0	0.1	0.0		0.0	0.2	
Non-interest current account deficit that stabilizes debt ratio	8.0	13.6	29.3			4.7	7.8	6.5	5.5	5.7	6.1		2.1	3.0	
Key macroeconomic assumptions															
Real GDP growth (in percent)	2.1	2.2	3.0	1.8	1.3	3.5	4.0	4.0	4.0	4.0	4.1	3.9	4.4	5.1	4.6
GDP deflator in US dollar terms (change in percent)	-0.6	9.8	-5.2	7.4	9.4	5.6	4.3	4.2	4.2	4.0	4.1	4.4	3.3	3.3	3.3
Effective interest rate (percent) 5/	0.8	0.8	0.8	0.6	0.2	0.5	0.8	1.2	1.4	1.6	2.0	1.3	2.3	2.8	2.4
Growth of exports of G&S (US dollar terms, in percent)	10.1	15.7	-10.3	9.3	15.0	14.3	9.9	10.5	9.1	7.3	7.1	9.7	6.6	8.0	7.0
Growth of imports of G&S (US dollar terms, in percent)	6.3	12.7	5.0	15.8	11.7	7.6	6.7	3.0	3.2	3.5	4.5	4.8	6.7	8.1	6.3
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	32.3	31.5	32.8	31.3	31.2	32.3	31.9	27.1	23.6	26.3
Government revenues (excluding grants, in percent of GDP)	14.3	16.1	19.3			15.2	15.0	15.2	15.4	15.6	15.7		16.1	15.5	16.1
Aid flows (in Billions of US dollars) 7/	0.1	0.0	0.1			0.2	0.1	0.1	0.1	0.1	0.1		0.1	0.3	
of which: Grants	0.1	0.0	0.1			0.2	0.1	0.1	0.1	0.1	0.1		0.1	0.3	
of which: Concessional loans	...	...	...			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			29.5	10.5	10.1	9.9	9.8	9.4		9.5	9.6	9.5
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			93.8	77.9	78.0	78.7	79.9	86.1		81.9	75.6	80.7
Memorandum items:															
Nominal GDP (Billions of US dollars)	0.5	0.6	0.6			0.7	0.7	0.8	0.8	0.9	1.0		1.4	3.1	
Nominal dollar GDP growth	1.5	12.2	-2.4			9.4	8.4	8.4	8.3	8.2	8.4	8.5	7.8	8.5	8.0
PV of PPG external debt (in Billions of US dollars)			0.1			0.1	0.1	0.1	0.1	0.2	0.2		0.3	0.8	
(PVt-PVt-1)/GDPt-1 (in percent)						2.5	3.4	3.3	2.9	2.7	1.6	2.7	2.1	2.8	2.2
Gross workers' remittances (Billions of US dollars)	0.1	0.1	0.2			0.2	0.2	0.2	0.2	0.2	0.2		0.2	0.4	
PV of PPG external debt (in percent of GDP + remittances)	...	...	6.7			8.0	10.0	11.7	13.1	14.3	14.5		16.7	21.3	
PV of PPG external debt (in percent of exports + remittances)	...	...	20.8			25.1	31.5	37.4	42.4	46.9	48.5		60.6	87.1	
Debt service of PPG external debt (in percent of exports + remittanc	...	...	3.9			0.3	0.7	1.0	1.8	2.1	2.3		3.3	5.9	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 1b.Comoros: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013-2033
(In percent)

	Projections							
	2013	2014	2015	2016	2017	2018	**2023**	2033
PV of debt-to GDP ratio								
Baseline	10	12	14	16	17	17	**20**	24
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	10	10	11	13	14	15	**28**	43
A2. New public sector loans on less favorable terms in 2013-2033 2	10	13	16	19	21	21	**26**	36
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	10	13	15	17	18	19	**21**	26
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	10	14	19	21	22	22	**23**	25
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	10	13	16	18	20	20	**22**	28
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	10	23	34	35	36	35	**35**	28
B5. Combination of B1-B4 using one-half standard deviation shocks	10	22	35	36	37	36	**36**	30
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	10	17	20	22	24	24	**28**	34
PV of debt-to-exports ratio								
Baseline	64	78	89	98	107	109	**131**	178
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	64	64	68	77	86	92	**186**	315
A2. New public sector loans on less favorable terms in 2013-2033 2	64	84	101	115	128	134	**173**	260
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	64	78	90	99	107	110	**132**	179
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	64	103	163	173	184	186	**212**	253
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	64	78	90	99	107	110	**132**	179
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	64	146	213	217	222	221	**232**	207
B5. Combination of B1-B4 using one-half standard deviation shocks	64	146	235	239	246	245	**259**	239
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	64	78	90	99	107	110	**132**	179
PV of debt-to-revenue ratio								
Baseline	66	82	94	103	110	111	**121**	157
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	66	67	72	81	88	93	**173**	279
A2. New public sector loans on less favorable terms in 2013-2033 2	66	88	107	121	133	135	**160**	230
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	66	85	101	111	119	119	**130**	169
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	66	93	126	133	139	138	**144**	164
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	66	88	107	118	125	126	**138**	179
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	66	154	226	228	229	224	**215**	184
B5. Combination of B1-B4 using one-half standard deviation shocks	66	148	230	234	236	231	**223**	197
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	66	116	133	146	156	156	**171**	222

Table 1C. Comoros: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013-2033

(In percent)

Debt service-to-exports ratio								
Baseline	1	2	2	4	5	5	**7**	12
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	1	2	2	3	4	4	**8**	18
A2. New public sector loans on less favorable terms in 2013-2033 2	1	2	2	4	5	6	**8**	18
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	1	2	2	4	5	5	**7**	12
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	1	2	4	7	8	8	**11**	18
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	1	2	2	4	5	5	**7**	12
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	1	2	4	8	8	9	**10**	18
B5. Combination of B1-B4 using one-half standard deviation shocks	1	2	5	9	9	10	**11**	20
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	1	2	2	4	5	5	**7**	12
Debt service-to-revenue ratio								
Baseline	1	2	3	4	5	5	**7**	11
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	1	2	2	4	4	5	**7**	16
A2. New public sector loans on less favorable terms in 2013-2033 2	1	2	2	5	6	7	**7**	16
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	1	2	3	5	5	6	**7**	11
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	1	2	3	5	6	6	**7**	12
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	1	2	3	5	5	6	**7**	12
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	1	2	5	8	9	9	**9**	16
B5. Combination of B1-B4 using one-half standard deviation shocks	1	2	5	9	9	9	**10**	16
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	1	3	4	6	7	7	**9**	15
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	26	26	26	26	26	26	**26**	26

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming) an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 2a.Comoros: Public Sector Debt Sustainability Framework, Baseline Scenario, 2010-2033

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate						Projections			
	2010	2011	2012	Average 5/	Standard Deviation 5/	2013	2014	2015	2016	2017	2018	2013-18 Average	2023	2033	2019-33 Average
Public sector debt 1/	51.4	47.2	21.2			22.4	25.7	27.9	29.3	30.6	30.4		31.3	35.3	
of which: foreign-currency denominated	49.2	44.7	18.9			20.2	22.7	25.1	26.7	27.9	27.7		28.6	32.6	
Change in public sector debt	-3.3	-4.3	-26.0			1.2	3.3	2.2	1.4	1.2	-0.2		0.4	0.7	
Identified debt-creating flows	-7.6	-7.4	-13.7			-19.9	-0.1	-0.2	-0.7	-1.0	-1.3		-1.9	-1.3	
Primary deficit	-7.4	-1.8	-3.9	-0.4	3.3	-18.3	1.5	1.5	1.1	0.8	0.6	-2.1	-0.2	0.6	0.0
Revenue and grants	29.2	23.6	28.9			43.7	24.1	23.9	24.1	24.2	24.4		24.8	24.2	
of which: grants	14.9	7.5	9.6			28.6	9.1	8.7	8.7	8.7	8.7		8.7	8.7	
Primary (noninterest) expenditure	21.7	21.8	25.0			25.4	25.6	25.4	25.2	25.1	24.9		24.6	24.7	
Automatic debt dynamics	-0.2	-5.7	1.3			-1.6	-1.6	-1.7	-1.8	-1.8	-1.8		-1.6	-1.9	
Contribution from interest rate/growth differential	-2.9	-3.0	-2.2			-1.3	-1.4	-1.5	-1.6	-1.6	-1.6		-1.6	-1.9	
of which: contribution from average real interest rate	-1.8	-1.9	-0.8			-0.5	-0.5	-0.5	-0.5	-0.5	-0.4		-0.3	-0.2	
of which: contribution from real GDP growth	-1.1	-1.1	-1.4			-0.7	-0.9	-1.0	-1.1	-1.1	-1.2		-1.3	-1.7	
Contribution from real exchange rate depreciation	2.8	-2.6	3.5			-0.3	-0.3	-0.2	-0.2	-0.2	-0.2		...	...	
Other identified debt-creating flows	0.0	0.0	-11.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	-11.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	4.3	3.2	-12.3			21.1	3.4	2.5	2.1	2.2	1.1		2.2	2.0	
Other Sustainability Indicators															
PV of public sector debt	2.2	2.5	10.7			12.2	15.3	17.2	18.6	19.8	20.0		22.3	27.1	
of which: foreign-currency denominated	0.0	0.0	8.4			10.0	12.3	14.4	16.0	17.2	17.3		19.6	24.4	
of which: external	...	...	8.4			10.0	12.3	14.4	16.0	17.2	17.3		19.6	24.4	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	-3.4	1.9	0.0			-16.1	3.8	4.6	4.4	4.0	3.8		3.3	4.7	
PV of public sector debt-to-revenue and grants ratio (in percent)	7.7	10.6	37.2			27.8	63.3	72.1	77.1	81.8	82.2		89.8	112.1	
PV of public sector debt-to-revenue ratio (in percent)	15.7	15.5	55.6			80.2	101.8	113.1	120.6	127.5	127.6		138.2	174.9	
of which: external 3/	...	...	43.6			65.9	82.1	94.4	103.5	110.5	110.6		121.3	157.5	
Debt service-to-revenue and grants ratio (in percent) 4/	6.1	6.9	5.5			0.2	1.2	1.6	2.7	3.1	3.4		4.3	6.8	
Debt service-to-revenue ratio (in percent) 4/	12.5	10.1	8.2			0.7	1.9	2.5	4.3	4.8	5.3		6.6	10.7	
Primary deficit that stabilizes the debt-to-GDP ratio	-4.2	2.5	22.1			-19.5	-1.8	-0.8	-0.3	-0.4	0.8		-0.6	-0.1	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	2.1	2.2	3.0	1.8	1.3	3.5	4.0	4.0	4.0	4.0	4.1	3.9	4.4	5.1	4.6
Average nominal interest rate on forex debt (in percent)	0.8	0.8	0.8	0.6	0.2	0.5	0.8	1.2	1.4	1.6	2.0	1.3	2.3	2.8	2.4
Average real interest rate on domestic debt (in percent)	-4.2	-4.5	-2.5	-3.6	1.4	-3.0	-3.1	-3.1	-3.1	-3.1	-3.2	-3.1	-3.2	-3.2	-3.2
Real exchange rate depreciation (in percent, + indicates depreciation	5.6	-5.7	8.2	-2.8	7.8	-2.0	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	4.4	4.7	2.6	3.8	1.5	3.1	3.2	3.2	3.2	3.2	3.3	3.2	3.3	3.3	3.3
Growth of real primary spending (deflated by GDP deflator, in percer	0.0	0.0	0.2	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0
Grant element of new external borrowing (in percent)	...	...	...	...	...	32.3	31.5	32.8	31.3	31.2	32.3	31.9	27.1	23.6	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2b.Comoros: Sensitivity Analysis for Key Indicators of Public Debt 2013-2033

	Projections							
	2013	2014	2015	2016	2017	2018	2023	2033
PV of Debt-to-GDP Ratio								
Baseline	12	15	17	19	20	20	22	27
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	12	14	15	16	17	17	22	38
A2. Primary balance is unchanged from 2013	12	15	17	18	20	20	26	37
A3. Permanently lower GDP growth 1/	12	15	17	19	20	21	24	34
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	12	16	20	22	24	25	30	40
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	12	16	19	21	22	22	24	28
B3. Combination of B1-B2 using one half standard deviation shocks	12	15	18	20	22	23	28	37
B4. One-time 30 percent real depreciation in 2014	12	19	20	21	22	22	23	28
B5. 10 percent of GDP increase in other debt-creating flows in 2014	12	23	24	25	26	26	28	30
PV of Debt-to-Revenue Ratio 2/								
Baseline	28	63	72	77	82	82	90	112
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	28	58	63	65	67	67	80	125
A2. Primary balance is unchanged from 2013	28	62	70	75	81	83	106	151
A3. Permanently lower GDP growth 1/	28	64	73	78	84	85	98	138
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	28	66	80	88	96	99	119	161
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	28	68	81	85	90	90	96	116
B3. Combination of B1-B2 using one half standard deviation shocks	28	64	73	81	88	91	110	149
B4. One-time 30 percent real depreciation in 2014	28	78	84	87	89	89	94	115
B5. 10 percent of GDP increase in other debt-creating flows in 2014	28	94	101	105	108	108	111	124
Debt Service-to-Revenue Ratio 2/								
Baseline	0	1	2	3	3	3	4	7
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	0	1	1	2	3	3	4	7
A2. Primary balance is unchanged from 2013	0	1	2	3	3	3	5	9
A3. Permanently lower GDP growth 1/	0	1	2	3	3	3	5	8
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	0	1	2	3	4	4	5	9
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	0	1	2	3	3	4	4	7
B3. Combination of B1-B2 using one half standard deviation shocks	0	1	2	3	3	4	5	9
B4. One-time 30 percent real depreciation in 2014	0	1	2	4	4	5	6	11
B5. 10 percent of GDP increase in other debt-creating flows in 2014	0	1	3	4	4	4	5	8

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.